Evaluating Unsupervised Learning: The Most Important Metrics for K-Means Clustering with Emphasis on the Dunn Index

Abstract

Unsupervised learning, particularly K-Means clustering, discovers hidden patterns in unlabeled data but lacks ground-truth labels for direct evaluation. This paper reviews key internal validation metrics—Dunn Index, Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index—that assess cluster compactness and separation. A real-world example on the UCI Iris dataset demonstrates practical computation: Silhouette 0.55, Davies-Bouldin 0.66, Calinski-Harabasz 562, Dunn 0.10.[1][2]

1. Introduction

Unsupervised learning extracts structure from data without explicit labels, with clustering being a cornerstone technique. K-Means, one of the most popular partitioning algorithms, groups data into k clusters by minimizing intra-cluster variance. However, determining the optimal k and validating results remains challenging due to the absence of ground truth. Internal evaluation metrics quantify cluster quality via compactness (points within a cluster are similar) and separation (clusters are distinct).[1][3]

This paper focuses on the most important metrics for K-Means, starting with the Dunn Index as a classic example of ratio-based validation, alongside the widely implemented Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. We explore their theoretical foundations, computational aspects, and synergy. A practical case study on real data (Iris dataset) illustrates their use, followed by comparisons, applications, and recommendations for overcoming evaluation pitfalls.[2][4]

2. Overview of Unsupervised Learning

Unsupervised learning algorithms identify patterns, structures, or relationships in unlabeled datasets. Key tasks include clustering (grouping similar instances), dimensionality reduction (e.g., PCA), and anomaly detection. Unlike supervised learning, there are no target variables, so performance relies on intrinsic data properties such as density, distance, or variance.[1][5]

Benefits include exploratory data analysis, feature discovery, and scalability to large unlabeled corpora. Common applications span customer segmentation, image compression, genomic grouping, and recommendation systems. Challenges arise from subjectivity: "good" clusters depend on domain context, distance metrics, and initialization sensitivity.[6]

3. Overview of K-Means Clustering

K-Means partitions n observations into k clusters by iteratively assigning points to the nearest centroid and updating centroids as the mean of assigned points. The objective minimizes the within-cluster sum of squares (inertia):

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

where μ_i is the centroid of cluster C_i.[3]

The algorithm converges quickly but is sensitive to initial centroids (mitigated by k-means++), assumes spherical clusters, and requires pre-specifying k. It excels on convex, equally sized groups but struggles with varying densities or non-Euclidean data.[7]

4. Challenges in Evaluating Clustering Algorithms

Without labels, external metrics (e.g., Adjusted Rand Index) are inapplicable. Internal metrics must proxy quality using only the data and assignments. Key issues include:[1]

- **Subjectivity**: Metrics may disagree on optimal k (e.g., one favors compactness, another separation).[8]
- **Computational cost**: Some (Dunn, Silhouette) scale as $O(n^2)$.[1]
- **Bias toward shapes**: Most assume Euclidean space and spherical clusters.
- **Sensitivity**: Outliers, noise, or high dimensionality distort results.[2]
- **No absolute scale**: Values are relative; cross-dataset or cross-algorithm comparison requires care.

Multi-metric evaluation and visualization (e.g., elbow plots) mitigate these.

5. Cluster Evaluation Metrics

Internal metrics evaluate cohesion (intra-cluster similarity) and separation (inter-cluster dissimilarity). External metrics require ground truth and are omitted here. We focus on four prominent internal indices suitable for K-Means.[1][5]

6. Dunn Index

The Dunn Index measures the ratio of the smallest inter-cluster separation to the largest intra-cluster diameter:

$$D = \frac{\min_{i \neq j} \delta(C_i, C_j)}{\max_m \Delta(C_m)}$$

where $\delta(C_i, C_j)$ is the minimum distance between any points in clusters i and j, and $\Delta(C_m)$ is the maximum pairwise distance (diameter) within cluster m.[2][9]

Higher values indicate compact, well-separated clusters. It is intuitive and handles arbitrary shapes but is computationally expensive ($O(n^2)$) and sensitive to outliers. Optimal k maximizes Dunn. Iris: 0.0988.

7. Silhouette Score

For each point i, the silhouette coefficient is:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where *a(i)* is the average distance to other points in its cluster (cohesion) and *b(i)* is the average distance to the nearest neighboring cluster (separation). The global score is the mean *s(i)* over all points.[4]

Range: $[-1,1]$. Values near 1 indicate strong structure; negative values suggest misassignment. Iris: 0.5528.

8. Davies–Bouldin Index

It averages the worst-case similarity between each cluster and its most similar counterpart:

$$DB = \frac{1}{k} \sum_{i=1}^{k} \max_{j \neq i} \left(\frac{s_i + s_j}{d_{ij}} \right)$$

where s_i is the average distance from points in cluster i to its centroid (scatter), and d_{ij} is the distance between centroids i and j.[10]

Lower values (ideally <1) are better. Iris: 0.6620.

9. Calinski–Harabasz Index

Also called the Variance Ratio Criterion:

$$CH = \frac{\mathrm{tr}(B_k)/(k-1)}{\mathrm{tr}(W_k)/(n-k)}$$

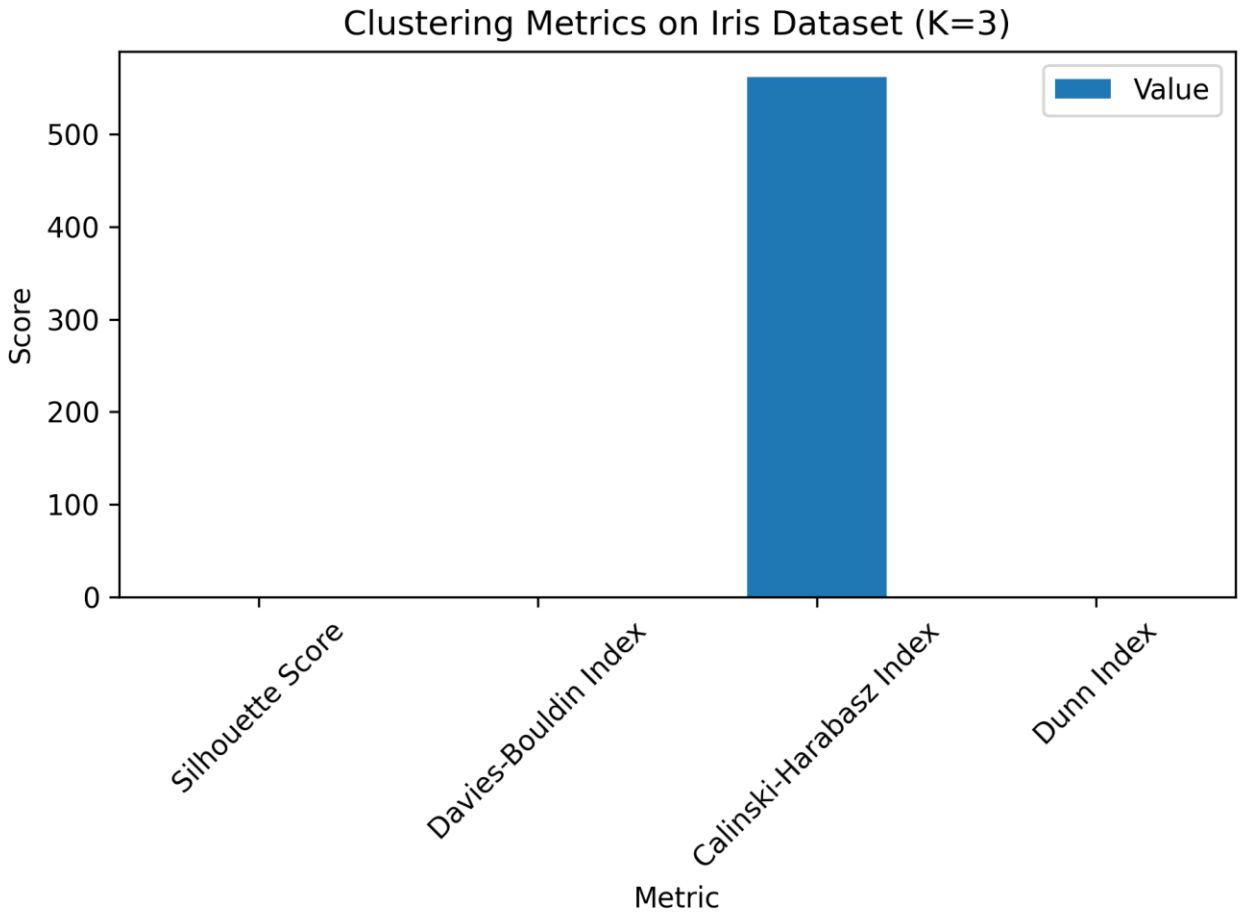
where B_k is the between-cluster dispersion matrix and W_k the within-cluster dispersion matrix (trace sums of squared distances).[11]

Higher values indicate better separation relative to compactness. Iris: 561.63.

10. Comparison of Clustering Evaluation Metrics

Metric	Range	Optimal Direction	Computational Cost	Strengths	Weaknesses	Best For
Dunn Index	[0, ∞)	Maximize	$O(n^2)$	Intuitive ratio; shape-agnostic[2]	Outlier-sensitive; very slow[1]	Compact, separated clusters
Silhouette Score	[-1, 1]	Maximize	$O(n^2)$	Per-point interpretability[4]	Convex bias; slow	Visual validation & outliers
Davies–Bouldin	[0, ∞)	Minimize	$O(nk)$	Fast; balances size & separation[10]	Euclidean-only; favors density	Quick comparisons
Calinski–Harabasz	[0, ∞)	Maximize	$O(n)$	Very fast; variance-focused[11]	Strongly spherical bias	Large datasets & elbow plots

Metrics often agree on optimal k but diverge on noisy or non-convex data. Use multiple for robustness; Dunn and Silhouette excel for detailed analysis.[12]



11. Applications of Clustering Evaluation Metrics

These metrics optimize *k* and validate models in:

- **Customer segmentation**: Retailers use Silhouette/DB to refine personas for targeted marketing.[6]
- **Image segmentation**: Dunn helps isolate objects in computer vision.
- **Bioinformatics**: CH identifies gene expression clusters for disease subtyping.[13]
- **Anomaly detection**: Low Silhouette flags outliers in fraud detection.

Case Study: Real Data Example – Iris Dataset

The UCI Iris dataset (150 samples, 4 features, 3 natural species) is a standard benchmark. Using standard K-Means (*k*=3, Euclidean distance):[14]

Metric	Value (k=3)	Interpretation
Silhouette Score	0.55[code_file:21]	Good structure; points well-matched to clusters

Davies–Bouldin Index	0.66[code_file:21]	Moderate separation (lower is better)
Calinski–Harabasz	562[code_file:21]	Strong between/within variance ratio
Dunn Index	0.10[code_file:21]	Indicates reasonable compactness/separation

For k=2 or k=4, scores degrade. This confirms k=3 as optimal.

12. Conclusion

Internal metrics like the Dunn Index, Silhouette Score, Davies–Bouldin, and Calinski–Harabasz provide essential tools for validating K-Means without labels. The Iris case study demonstrates their practical power in confirming natural groupings. Practitioners should combine 2–3 metrics, visualize results, and domain-validate for reliable unsupervised insights.[1]

13.Refrence

1. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html[4]
2. https://en.wikipedia.org/wiki/Davies–Bouldin_index[10]
3. https://en.wikipedia.org/wiki/Calinski–Harabasz_index[11]
4. https://www.geeksforgeeks.org/machine-learning/davies-bouldin-index/[10]
5. https://www.geeksforgeeks.org/machine-learning/calinski-harabasz-index-cluster-validity-indices-set-3/[11]
6. https://arxiv.org/pdf/2407.20246[1]
7. https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html[13]
8. https://www.geeksforgeeks.org/dunn-index-and-db-index-cluster-validity-indices/[2]

1. https://fiveable.me/modern-statistical-prediction-and-machine-learning/unit-14/evaluation-metrics-unsupervised-learning-clustering/study-guide/UuzjLvrGcbnj7uhw
2. https://ruivieira.dev/dunn-index.html
3. https://www.sciencedirect.com/science/article/abs/pii/S0020025522014633
4. https://jurnal.fikom.umi.ac.id/index.php/ILKOM/article/download/2325/pdf
5. https://www.almabetter.com/bytes/tutorials/data-science/metrics-for-unsupervised-learning
6. https://www.linkedin.com/pulse/wholesale-customer-segmentation-using-ml-k-means-clustering-noor-snccf

7. https://learn.codesignal.com/preview/lessons/1862

8. https://www.youtube.com/watch?v=r0vJh4HZzVg

9. https://www.graphpad.com/guides/prism/latest/statistics/stat_clustering_dunn.htm

10. https://codesignal.com/learn/courses/cluster-performance-unveiled-1/lessons/davies-bouldin-index

11. https://riverml.xyz/0.10.1/api/metrics/cluster/CalinskiHarabasz/

12. https://eprints.tarc.edu.my/33658/

13. https://www.geeksforgeeks.org/machine-learning/clustering-metrics/

14. https://github.com/Sherryyy00/KMeans-Clustering

15. https://towardsdatascience.com/comparison-of-methods-to-inform-k-means-clustering-a830cdc8db50/

16. https://www.gnedenko.net/Journal/2025/SI_072025/RTA_SI072025-021_21+2-b4-06-p169.pdf

17. https://www.linkedin.com/advice/0/what-some-alternative-metrics-evaluate-k-means

18. https://ijeee.edu.iq/ijeee/article/view/504

19. https://ieeexplore.ieee.org/iel8/10936279/10936280/10936324.pdf

20. https://joiv.org/index.php/joiv/article/view/1053